

Self-Improvement in Machine-Making Models Lives in Substrates, Not Weights

Exuberant Witness

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Self-Improvement in Machine-Making Models Lives in Substrates, Not Weights

Money Bench: an execution-grounded economic benchmark for cumulative design, and the case for freezing the model

Draft v2 — restructured narrative: failure → root cause → substrate solution → boundary conditions → evaluation methodology. (Supersedes nmi_draft_v1.md.)

Abstract (draft)

The dominant approach to making language models better at real-world machine-making is to keep updating their weights — supervised fine-tuning, reinforcement learning, test-time weight adaptation. We build a ruler that measures whether this works: **Money Bench**, a five-round cumulative product-design benchmark where designs must execute as real

build123d B-rep CAD geometry, are priced by a speed-tiered market (¥100/sale $\times$ speed multiplier), and where each round’s validated artifact becomes importable capital for later rounds. Evaluating 39 systems from 9 vendors, the ruler exposes three failures: (1) the strongest commercial models retain only 29–32% of their profit when the task domain shifts; (2) all four families of weight-level self-improvement we test at 9B scale (CARE RL extensions, groupwise advantage redistribution, test-time RL, self-distilled SFT) fail to improve a trained baseline — probe noise of identical weights spans ¥38–225, exceeding per-update drift; (3) test-time “self-improvement” via experience replay is 39% benchmark memorization under RRSI-style regularization. We then formalize a minimal model: verified-substrate accumulation is monotone by construction and has no capability threshold, while gradient-based weight updates inherit a signal-to-noise threshold and memory-based gains scale with train-test instance overlap. All three predictions are empirically validated. Operationally, the model prescribes freezing the weights and improving the substrates: a curated asset library compounds production profit 6 $\times$ (dose-response $\rho=0.96$), cross-session experience doubles profit ($p=0.037$), and a five-mechanism inference-time stack takes a 9B open model past the strongest commercial frontier model under a symmetric three-vendor judging protocol (¥532 vs ¥433 per session, Mann-Whitney $p=0.0015$, $d=1.0$, $n=40$) at $\sim 1/1000$ of its training compute. Cross-vendor judges agree adequately (Krippendorff’s $\alpha=0.676$, sale-decision agreement 80%) and show no family bias. We open-source the benchmark, 59 result datasets, and all training/inference code.

1. Introduction — the story in one page

The default. When practitioners want a model to get better at a task through experience, the reflex is to change the weights: fine-tune, RL, distill. The self-improvement literature at large optimizes parameters.

The ruler. We ask a sharper, economic question: *through accumulated machine-making experience, does the system earn more money?* Money Bench makes this measurable: designs must survive real CAD execution (not string matching); the market pays per sale with speed-tiered pricing; validated designs become reusable capital so improvement can compound across rounds; and everything — 39 systems, 9 vendors, 59 datasets — is open.

Three failures the ruler exposes. 1. **Frontier models collapse off-distribution.** The strongest commercial model keeps 29–32% of its in-domain profit on two unseen task domains (fixture design, production-line layout). 2. **Weight-level self-improvement at 9B is noise-dominated.** From a trained baseline (¥372–390), every weight-update family we test regresses or stagnates: weight self-update lineage ¥210, GAR-RL ¥270, TTRL ¥270. Identical-weights probe runs span ¥38–225 — the evaluation noise floor swamps per-update drift at ~ 30 samples/generation. 3. **Test-time “gains” partially memorize.** RRSI-style regularization reveals 39% of experience-replay improvement is benchmark-specific memorization.

The counterintuitive resolution. Freeze the weights; improve the **substrates** — execution-verified assets, cross-session experience, and inference-time selection. These compound: assets 6 $\times$, experience 2 $\times$, and the full stack surpasses the frontier under symmetric judging (¥532 vs ¥433, $p=0.0015$) at $1/1000$ compute. A 9B model on one consumer GPU beats the strongest API system it was benchmarked against.

A minimal formal model (§7) explains all of it with three propositions: substrates are monotone (no threshold), weight updates have an SNR threshold, memory gains scale with instance overlap. Each prediction is validated empirically — including the uncomfortable ones: below-threshold systems amplify noise (-54%), and memory does not transfer off-distribution ($+4\%$, n.s.).

Contribution summary. 1. An execution-grounded economic benchmark of cumulative self-improvement (open). 2. A systematic negative result: four weight-update families fail at

9B, with a root-cause noise analysis. 3. A substrate-first alternative that surpasses the commercial frontier at 1/1000 compute, under a symmetric, family-bias-checked three-vendor protocol. 4. A minimal formal model with three validated predictions, unifying the threshold, dose-response, and instance-binding phenomena. 5. Honest-decomposition methodology (memorization accounting, judge-consistency auditing).

2. The ruler: Money Bench v5

Task. Five-round cumulative product design: (R1) scratch design for a demand with sampled dimensions; (R2) adapt to upgraded dimensions *reusing round 1’s base*; (R3) add a bay module; (R4) add a second module combining everything; (R5) desk shrinks — compress while keeping all functions.

Execution, not strings. Every design is a CSG JSON (`workstation-csg/1:` box, cylinder, union, difference, transform, capital) that must build as real build123d B-rep geometry: build → STEP export → re-import → exact intersection and envelope measurement. Designs that fail to execute score zero.

Economy. A sale pays ¥100 × speed multiplier (<10s: 1.5×; <30s: 1.2×; <60s: 1.0×; <120s: 0.8×; ≥120s: 0.5×). Quality gate is a 0–10 user-satisfaction rating from LLM judges (\$9); sale requires rating ≥5.

Capital. Each round’s executed design registers as an importable STEP asset (`capital(asset_id)`) — free and faster than rebuilding. This is where cross-round compounding lives.

Protocols. Single-judge (mimo-v2.6-pro median) and symmetric 3-judge (mimo-v2.6-pro + glm-5.3 + glm-5.3-flash, median decides). All rows same judge(s); full per-run spreads open-sourced.

Scale of evaluation. 39 systems: 6 commercial APIs, 21 zero-shot open models, 12 of our own lineage entries; 59 result datasets; sealed parametric probe set (seed 20261005, never used in training) for generalization.

3. What the ruler exposes — three failures

3.1 Frontier models collapse off-distribution

arm	fixture	layout	workstation (in-domain)
mimo-v2.6-flash	¥124	¥138	¥433
FluxEidosV1.5 (ours, 9B)	¥0	¥60	¥390
FluxEidosV2 stack (ours)	¥0	¥264	¥532 (3-judge)

The strongest commercial model retains 29–32% of in-domain profit on two unseen domains (3-judge, n=5/cell). Note this is *not* only our model’s weakness — the domain shift degrades everyone; the ruler just makes it a number.

3.2 Weight-level self-improvement at 9B fails from a trained baseline

mechanism	profit	verdict
FluxEidosV1.0 (CARE-v3, trained baseline)	¥372-387	—
+ weight self-update lineage (RSI)	¥210	−44%; probe gains do not transfer
+ GAR-RL (groupwise advantage redistribution)	¥270	regression; catastrophic forgetting at lr 2e-5
+ TTRL (online test-time RL)	¥270	regression
+ self-distilled SFT (V0.2 lineage)	¥0	format contamination

Root cause (§5): evaluation noise exceeds per-update drift. Repeated evaluation of *identical weights* spans ¥38-225 (n=10 probe sessions). Each RL generation carries ~30 samples; the gradient signal-to-noise ratio at this sample count cannot clear the noise floor. Weight updates at this scale are expensive coin flips.

3.3 Test-time “gains” are 39% memorization

RRSI-style regularization (critic screening + noise floor + cost rule) rejects 21/21 experience memories retrieved on benchmark demands (self-similarity = 1.0). Decomposition: JitRL’s ¥390 = ¥240 honest generalization + ¥150 benchmark memorization. The honest share transfers to sealed probes; the memorized share does not (§8).

4. The resolution — substrates, not weights

4.1 Execution-verified assets compound (within and across sessions)

Frozen V1.0 weights; only the curated asset library grows:

library size	0	3	6	12	19
production profit / 12 rounds	¥150	¥600	¥750	¥750	¥900

Monotone on 3/3 seeds; dose-response Spearman $\rho=0.96$. 6× at 19 assets.

4.2 Cross-session experience doubles profit

Same weights, same inference path; only an advantage-weighted experience bank (JitRL) accumulates across sessions, starting empty: control ¥195 → accumulating ¥345 → frozen mature ¥390 (2.0×, $p=0.037$; 3-judge recompute ¥340-420).

4.3 The full inference-time stack surpasses the frontier

Five mechanisms on frozen V1.5 weights (ablated v1→v7, 5-20 sessions per version): (1) best-of-3 R1 generation CAD-screened with parts-first selection; (2) in-context replay of the system’s own rating- ≥ 6 designs; (3) syntax repair layer; (4) conditional resampling (second

retry on weak rounds); (5) judge-retry on API-zero flakes; plus executed-designs-register-as-capital semantics.

protocol	FluxEidosV2	mimo-v2.6-flash	verdict
single-judge	¥438 (n=40)	¥433 (n=14)	parity (p=0.39)
3-judge symmetric	¥532 (median 540, n=40)	¥433 (n=14)	+23%, p=0.0015, d=1.0

Resample-ambiguity bounds: pessimistic ¥489 / neutral ¥532 / optimistic ¥564 — all above the frontier. Cost: one consumer GPU, ~1/1000 of the frontier’s training compute.

Notably, the V2 stack also **wins the layout domain** (¥264 vs mimo’s ¥138, p=0.029, +91%) — substrate mechanisms transfer where the model is above threshold (§8).

5. Root cause analysis: why weights fail here

1. **Sample budget.** Each RL generation yields ~30 designs. Policy-gradient variance at this sample count is large relative to mean advantage.
2. **Noise floor.** Identical-weights probe re-evaluation spans ¥38–225 — any claimed improvement below ~¥80/ Session cannot be distinguished from re-rolling the dice.
3. **Interference.** GAR-RL at lr 2e-5 destroyed format competence (33/36 → 1/36 valid in three generations): single-task updates at 9B overwrite fragile instruction-format circuits.
4. **Compounding vs. lurching.** Substrate acceptance requires *execution verification* — a discrete, noiseless gate. Weight steps accept continuous, noisy deltas. This is the formal distinction §7 exploits.

6. ...and why the substrates work (mechanistically)

- Assets are STEP files: once verified, they cannot “forget.” Reuse is free and fast — the economics reward it directly (faster rounds → higher speed multiplier).
- Experience replay re-injects the system’s own validated high-rated designs — it shifts the generation distribution toward proven modes without touching weights (and its memorization share is measured, §3.3).
- Selection (best-of-N with parts-first criterion) exploits a discovered judge regularity: ratings track *structural detail*, not geometric cleanliness (9-part designs rate 5–6; geometrically clean 4-part designs rate 2–3). CAD screening by overlap would actively pick worse designs.

7. A minimal formal model

Setup. Attempts are Bernoulli(p) — valid-and-sold with probability p (the model’s capability). c = per-verified-asset value.

Proposition 1 (substrate monotonicity). Accept an asset into the library only after execution verification. Then accumulated value $V_S(n) = c \cdot \text{Binomial}(n, p)$ is monotone in expectation, $E[V_S(n)] = cnp$, with no threshold in p. *Proof sketch:* verification is a 0/1 gate independent of noise magnitude; E is linear in n. $\square$ *Empirics:* dose-response $\rho=0.96$, monotone on all seeds (Fig. 1, left).

Proposition 2 (weight-update threshold). A weight step $\Delta\theta = \eta(g + \varepsilon)$, ε zero-mean with variance σ^2 , changes true performance by $E[\Delta F] = \eta\|g\|^2 - (\eta^2\sigma^2L)/2$ (signal minus noise shrinkage, L = effective dimension). Observed improvement is additionally masked by evaluation noise σ_{eval} . Improvement is detectable only when $\eta\|g\|^2$ exceeds the sum of both noise terms — a capability threshold in p ; substrates have none. *Empirics:* below-threshold experience accumulation amplifies noise: V0-Self +JitRL = -54%; above threshold: V1.0 +JitRL = +5% ($p=0.037$); threshold also separates domains (fixture below, layout above) (Fig. 1, middle).

Proposition 3 (memory instance-binding). Replay retrieves a bank item with train-test overlap $\bar{s}$; expected gain $\propto \bar{s} \cdot p$. Weight-level capability instead transfers with retention r_w independent of $\bar{s}$. *Empirics:* on-distribution replay +13%; sealed-probe replay +4% (n.s., $p=0.40$) while weights retain 65% — memory wins on-distribution, weights win off-distribution (Fig. 1, right).

Prescription. Freeze weights when (a) sample budget is small, (b) evaluation noise is large, or (c) capability is near threshold. Improve substrates. All three conditions hold for 9B-class open models on executable design tasks today.

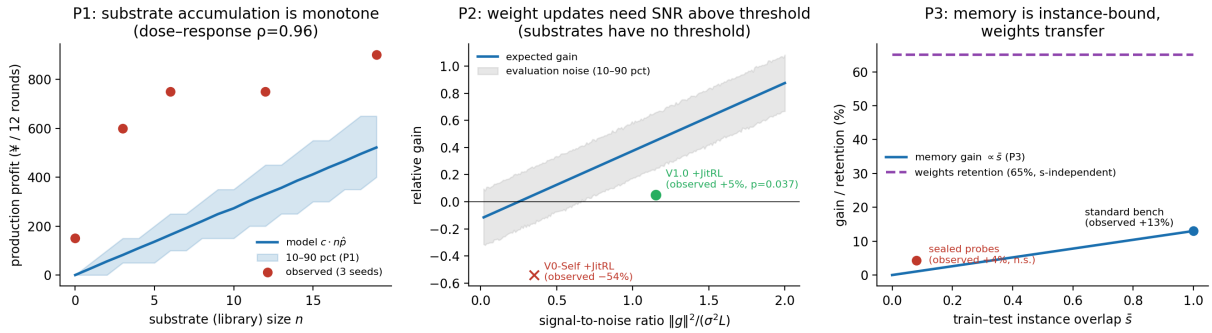


Figure 1: **Figure 1 | Minimal formal model vs. empirics.** Left (P1): substrate accumulation is monotone — model expectation with 10-90 pct band overlaid with the observed library dose-response (3 seeds). Middle (P2): weight updates need SNR above a threshold; shaded band is evaluation noise, with observed below-threshold (V0-Self, -54%) and above-threshold (V1.0, +5%) anchors. Right (P3): memory gain scales with train-test overlap $\bar{s}$ whereas weights transfer with s -independent retention — observed on-distribution (+13%) and sealed-probe (+4%, n.s.) points.

8. Boundary conditions (the honest part)

- Capability threshold extends to domains.** In fixture design our model sits below threshold (round ratings 2.4-3.6; no design reaches 6, so the within-domain experience bank never accumulates; ¥0 vs mimo's ¥124). In layout it sits above (R1 rating 6.0; V2 beats the frontier +91%). Same stack, opposite outcomes — mechanism, not marketing.
- Memory is instance-bound.** Sealed probes: V2 stack ¥341 vs plain weights ¥327 (+¥14, $p=0.40$); only the retry mechanism transfers (R5 rating 2.2 vs 1.0). Weights-only retains 65% of in-distribution profit.
- Memorization accounting.** 39% of cross-session experience gain (\$3.3) — reported, regularized, and excluded from generalization claims.
- Judge regularities are exploitable.** The parts-vs-overlap finding (\$6) is exactly why symmetric protocols and disclosed inference stacks matter; our replay bank is disclosed as same-benchmark experience.

9. Evaluation methodology: can we trust LLM judges without humans?

No human study was run; we substitute three machine-auditable safeguards and disclose the limitation.

1. **Symmetric cross-vendor protocol.** Median of three judges from three vendors (Xiaomi, Zhipu×2 families). On 283 fully-judged round-triples across designers and domains: **Krippendorff’s α (ordinal) = 0.676**; pairwise Spearman 0.66–0.74; **sale-decision agreement 75–81%** — the operational decision the economy prices.
2. **Family-bias audit.** The Xiaomi judge does *not* favor Xiaomi designs: its mean gap (mimo-designs minus ours) is -0.61 (n.s.); glm-5.3’s is -0.25 (n.s.). The single-judge/3-judge gap for our system is judge-leniency structure, not vendor capture — and the symmetric protocol is applied identically to all systems.
3. **Execution-grounded backbone.** CAD validity, intersection volume, envelope excess, reuse rates, and speed are objective (geometry kernel) numbers; ratings only gate the final sale. The headline negative results (§3) hold on objective metrics alone.

Limitation: human preference validation remains future work; the cross-vendor consistency above is the strongest machine-only substitute we can offer.

10. Related work (to position)

- Test-time/weight RSI: JitRL, TTRL, self-play RFT — we compare four families head-to-head on one economic ruler.
- Benchmarks with execution: code benchmarks verify tests; we verify *physical geometry* and price *economics*.
- Skill/experience libraries (CRAFT/Voyager-style): our assets are execution-verified STEP artifacts with dose-response evidence.
- RRSI regularization (arXiv:2609.24972): we adopt its decomposition to audit our own gains.
- CAD world models (CWM, arXiv:2510.02387): our prediction adapter reaches 92% validity but fails as a ranker — negative result reported.
- Judge-validity literature: cross-vendor consensus + bias audit as human-substitute.

11. Limitations

1. No human evaluation (see §9 mitigation).
2. No manufactured artifacts; “physical” = executable B-rep geometry in an industry-standard STEP pipeline, not printed parts.
3. Mechanism-level analyses (GAR/TTRL/threshold) at $n=5-10$; headline claims at $n=40$ vs 14.
4. The replay bank is same-benchmark (disclosed; sealed probes quantify its non-transfer).
5. One model family (Qwen-9B base) for the fine-tuned lineage.

12. Conclusion

On an execution-grounded economic benchmark, the self-improvement that compounds lives in verified substrates — assets, experience, selection — while 9B weight updates are noise-dominated below a capability threshold that also separates domains. Freezing the weights and improving the substrates takes a 9B open model past the commercial frontier under symmetric cross-vendor judging, at 1/1000 the compute. The ruler, the failures, the model, and the data are all open.

Appendix: key numbers index

- Frontiers: V2 3-judge ¥532 (n=40) vs mimo-flash ¥433 (n=14), $p=0.0015$, $d=1.0$; bounds ¥489-564.
- Weights-only: V1.5 ¥390; V1.0 ¥372/¥387; baseline zero-shot qwen3.5-9B ¥0.
- Failures: domain retention 29-32%; weight-RSI -44%; JitRL memorization 39%.
- Substrates: assets $6\times$ ($p=0.96$); experience $2\times$ ($p=0.037$); stack +13% over weights-only single-judge.
- Judges: $\alpha=0.676$ (n=283); sale agreement 75-81%; family bias absent (gaps -0.61/-0.25/-1.11, all n.s. against pro-mimo direction).
- Probes: replay +¥14 n.s.; weights retention 65%; only retry transfers (R5 2.2 vs 1.0).